

DAP Output Stage Rev1

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Attribution

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- **Drafter/documenter:** Claude, writing from Alexander's direction and source data

Part Identification

The output stage drives the headphones from the summing-stage outputs. Each channel is a unity-gain non-inverting follower with output-inclusive compensation.

Signal chain per channel: I/V ($\times 2$) $\rightarrow$ differential summing + MFB low-pass $\rightarrow$ **OPA1622 follower** $\rightarrow$ 3.5 mm jack.

- **Amplifier:** TI OPA1622IDRCR, dual, 10-pin VSON (DRC)
- **Jack:** Same Sky SJ1-3535NS, 3.5 mm stereo TRS, right angle, through hole, shielded, tip and ring switches
- **ESD:** TI ESD122DMXR, 2-channel bidirectional, on tip and ring

Design Requirements

- **Gain:** 1. The summing stage already sets 0.85 V peak (0.6 V RMS) and in-phase polarity.
- **Load:** Simgot EW300, $28\ \Omega \pm 15\%$ ($23.8\ \Omega$ worst case); ≈ 36 mA peak per channel at 0.85 V peak.
- **Output impedance:** $< 1\ \Omega$, 20 Hz to 20 kHz.
- **Coupling:** DC coupled.
- **Stability:** stable into cable capacitance.
- **No additional low-pass.** The I/V pole and MFB filter already provide ≈ 60 dB at 1 MHz; another filter would add passband loss, raise output impedance, and complicate stability.

Design Constraints

- ± 3.3 V supply rails
- Low noise
- Small footprint
- C0G capacitors in the signal path

Topology Rationale

- **Non-inverting follower over inverting stage:** keeps polarity, draws essentially no current from the MFB output, and applies the full loop gain as feedback for the lowest output impedance.
- **Output-inclusive compensation:** a small isolation resistor R_{iso} separates the op-amp from cable capacitance. Audio-frequency feedback (R_{fb}) is taken after R_{iso} , at the jack, so the loop corrects R_{iso} 's drop. High-frequency feedback (C_f) is taken before R_{iso} , so the loop does not see the capacitive load.

Pin Connections

OPA1622IDRCR

Pin	Name	Connection
1	+INA	SUMM_R
10	-INA	R_{fb} (to ring node) and C_f (to OUTA)
9	OUTA	$R_{iso} \rightarrow$ ring node (RIGHT_EAR)
5	+INB	SUMM_L
6	-INB	R_{fb} (to tip node) and C_f (to OUTB)
7	OUTB	$R_{iso} \rightarrow$ tip node (LEFT_EAR)
2	V+	+3V3_ANALOG, 0.1 μ F + 1 μ F to GND
4	V-	-3V3, 0.1 μ F + 1 μ F to GND
3	GND	GND (reference for EN)
8	EN	EN network (see below)

Pin	Name	Connection
Pad	Thermal pad	V- (-3.3 V), own pour and via field

SJ1-3535NS

Pin	Function	Connection
1	Sleeve	GND
2	Tip	Left output node (TIP), ESD122 IO2, 10 k Ω bleed to GND
3	Ring	Right output node (RING), ESD122 IO1
4	Tip switch (normally closed to tip, opens on insertion)	Detect node: 100 k Ω pull-up to +3V3_DIG, 10 k Ω series to PLUG_DET (GPIO)
5	Ring switch	Not connected
SH1-SH3	Shield	GND

Component Values

Ref	Value	Qty	Notes
R_iso	4.7 Ω	2	Op-amp output $\rightarrow$ jack node
R_fb	499 Ω , 0.1% (YAGEO RT0402BRD07499RL)	2	Jack node $\rightarrow$ -IN; same part as I/V R_f
C_f	1.5 nF C0G, 1% (Murata GCM1885C1H152FA16D)	2	Op-amp output $\rightarrow$ -IN; same part as I/V C_f
Rail decoupling	0.1 μ F X7R + 1 μ F X5R per rail	4	V+ and V-
EN series	54.9 k Ω	1	EN_SPEAKER $\rightarrow$ EN
EN pull-down	50 k Ω	1	Holds EN low at boot

Ref	Value	Qty	Notes
EN ramp	0.1 μ F X7R	1	EN node $\rightarrow$ GND
Detect pull-up	100 k Ω to +3V3_DIG	1	On the jack-side detect node
Detect series	10 k Ω	1	Detect node $\rightarrow$ PLUG_DET (GPIO); limits current when the tip swings negative
Tip bleed	10 k Ω	1	TIP $\rightarrow$ GND; holds the tip near 0 V when the amp is disabled
ESD	ESD122DMXR	1	Tip and ring, at the jack

Enable (EN)

- Logic threshold 0.82 V, referenced to the GND pin.
- $V_{EN} = 3.3 \text{ V} \times 50\text{k} / (54.9\text{k} + 50\text{k}) \approx 1.57 \text{ V}$ from a 3.3 V GPIO.
- Ramp time constant $(54.9\text{k} \parallel 50\text{k}) \times 0.1 \mu\text{F} \approx 2.6 \text{ ms}$.
- GPIO control is used instead of the datasheet's supply-derived divider (Figure 48) so firmware can mute on demand. Firmware enables the amp only after both analog rails are up, and keeps it disabled while unplugged.

Plug Detect

State	Detect node / GPIO	Reads
Unplugged, amp off (tip held by bleed)	$3.3 \text{ V} \times 10\text{k} / 110\text{k} \approx 0.3 \text{ V}$	Low
Unplugged, amp on (tip $\approx 0 \text{ V}$)	$\approx 0 \text{ V}$	Low
Plugged (switch open)	3.3 V	High

The GPIO draws no current, so the 10 k Ω series resistor does not shift the reading.

Equations and Results

- **Feedback handover:** $f_x = 1 / (2\pi \times R_{fb} \times C_f) = 1 / (2\pi \times 499 \Omega \times 1.5 \text{ nF}) \approx 213 \text{ kHz}$

- **Output impedance:** $Z_{out} \approx R_{iso} \times |R_{fb} / (R_{fb} + Z_{Cf})|$, with $Z_{Cf} = 1 / (2\pi f C_f)$

Quantity	Value ($R_{iso} = 4.7 \Omega$)
Z_{out} at 20 kHz	$\approx 0.40 \Omega$
Extra swing across R_{iso} at 36 mA peak	$\approx 0.17 V$
R_{fb} noise over 20 kHz	$\approx 2.9 \text{ nV}/\sqrt{\text{Hz}}$, $\approx 0.4 \mu\text{V RMS}$

Stability

Datasheet (phase margin vs capacitive load, $G = +1$): $\approx 70^\circ$ at 200 pF, $\approx 45^\circ$ at ≈ 430 pF, $\approx 30^\circ$ at 600 pF, $\approx 0^\circ$ at 1 nF.

Simulation: 10 mV step into the follower, C_{load} swept at 300 pF, 600 pF, 1 nF.

Load	Network	Overshoot (300 / 600 pF / 1 nF)
$23.8 \Omega \parallel C_{load}$	$2 \Omega / 499 \Omega / 1.5 \text{ nF}$	Clean at all three
$10 \text{ k}\Omega \parallel C_{load}$	Plain follower	Much worse than the compensated network
$10 \text{ k}\Omega \parallel C_{load}$	$2 \Omega / 499 \Omega / 1.5 \text{ nF}$	$\approx 10\% / 20\% / 40\%$
$10 \text{ k}\Omega \parallel C_{load}$	$4.7 \Omega / 499 \Omega / 1.5 \text{ nF}$	$\approx 10\% / 15\text{--}20\% / 30\%$, settles in 1–2 cycles

The 23.8Ω IEM load damps cable capacitance. $10 \text{ k}\Omega \parallel 1 \text{ nF}$ is a harsh bound (line-out into a long cable), outside the intended use.

Open Problems

- ☐ **Thermal pad pin.** Add to symbol and footprint, tied to $-3V3$.
- ☐ **Detect net label.** The jack-side detect node must not share the PLUG_DET name with the GPIO port, or the $10 \text{ k}\Omega$ series resistor is bypassed.
- ☐ **Output impedance simulation.** + input grounded, AC current into the jack node; confirm $\approx 0.40 \Omega$ at 20 kHz.
- ☐ **EN maximum voltage.** Confirm from the datasheet.
- ☐ **DC-fault protection and mute topology.**

Rev Log

Rev	Date	Change
1	2026-09-25	R_iso placed. Plug detect finalized: 100 k Ω jack-side pull-up, 10 k Ω series to GPIO, 10 k Ω tip bleed; Plug Detect section added. R_iso and plug-detect open problems closed; detect net label issue added.
0	2026-09-25	Initial release. Topology, pin connections, component values, EN network, equations, stability data, open problems.

Sources

- OPA1622 datasheet (TI): phase margin vs capacitive load; output transients during power up and power down
- SJ1-3535NS datasheet (Same Sky)
- [ESD122 product page, TI](#)